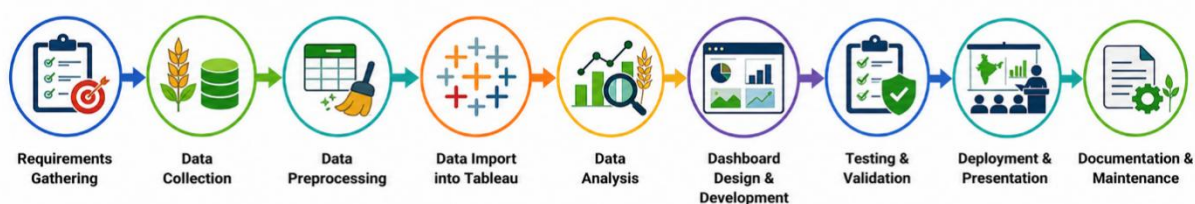


PROJECT DEVELOPMENT PHASES

The development of the India's Agricultural Crop Production Analysis (1997–2021) project follows a systematic approach to transform raw agricultural data into meaningful visual insights using Tableau. Each phase focuses on collecting, preparing, analyzing, and visualizing crop production data to identify production trends, seasonal patterns, state-wise performance, and yield variations. This structured process ensures accurate analysis and supports data-driven decision-making for agricultural planning and resource management.

They are different types of project development phases



1. Requirements Gathering

Definition: Requirements Gathering is the process of identifying the project objectives, user requirements, and expected outcomes before analysis begins.

- **In this phase,** the objectives of analyzing India's agricultural crop production from 1997 to 2021 are defined. The project scope, user requirements, and key performance indicators (KPIs) are identified to support meaningful agricultural analysis.

Key Insights:

- Defined project objectives and scope.
- Identified user requirements.
- Determined KPIs for agricultural analysis.

2. Data Collection

Definition:

Data Collection is the process of gathering relevant and reliable data required for analysis.

- **In this phase,** the agricultural crop production dataset (1997–2021) is collected from reliable sources. The dataset includes information such as state, crop, season, area cultivated, production, and yield.

Key Insights:

- Collected agricultural production data.
- Verified dataset completeness.
- Prepared data for preprocessing.

3. Data Preprocessing

Definition: Data Preprocessing is the process of cleaning and organizing raw data for analysis.

- **In this phase,** missing values, duplicate records, and inconsistencies are handled. The dataset is formatted and standardized to ensure accurate analysis.

Key Insights:

- Improved data quality.
- Removed inconsistencies.
- Prepared clean data for visualization.

4. Data Import into Tableau

Definition: Data Import is the process of loading the prepared dataset into Tableau Desktop for visualization and analysis.

- **In this phase,** the cleaned agricultural dataset is imported into Tableau. Data fields and data types are verified to ensure accurate visualization.

Key Insights:

- Connected the dataset to Tableau.
- Verified data fields and data types.
- Prepared data for dashboard development.

5. Data Analysis

Definition: Data Analysis is the process of examining data to identify trends, patterns, and meaningful insights.

- **In this phase,** crop production trends, seasonal variations, state-wise production, and yield patterns are analyzed using Tableau visualizations.

Key Insights:

- Identified crop production trends.
- Compared state-wise agricultural performance.
- Analyzed seasonal production patterns.

6. Dashboard Design and Development

Definition: Dashboard Design and Development is the process of creating interactive visualizations for effective data presentation.

- **In this phase**, interactive dashboards are designed using maps, charts, graphs, KPIs, and filters to visualize agricultural production across India.

Key Insights:

- Developed interactive dashboards.
- Created clear agricultural visualizations.
- Improved data interpretation.

7. Testing and Validation

Definition: Testing and Validation is the process of verifying the accuracy and functionality of the dashboard.

- **In this phase**, all dashboards, filters, and calculations are tested to ensure accurate visualization and reliable analytical results.

Key Insights:

- Verified dashboard accuracy.
- Validated calculations.
- Ensured reliable analysis.

8. Deployment and Presentation

Definition: Deployment and Presentation is the process of publishing and presenting the completed dashboard.

- **In this phase**, the dashboard is published and presented to demonstrate agricultural production trends and support informed decision-making.

Key Insights:

- Published the dashboard.
- Presented analytical findings.
- Supported agricultural decision-making.

9. Documentation and Maintenance

Definition:

Documentation and Maintenance is the process of recording project details and updating the dashboard when new data becomes available.

- **In this phase**, project documentation is completed, and the dashboard is maintained by updating agricultural datasets and improving visualizations as needed.

Key Insights:

- Completed project documentation.
- Updated dashboards regularly.
- Maintained accurate agricultural analysis.